Population Analysis

Text files

`StoreResults(data.files.analysis_dir, ['beh.dat'], [], data.dataname, {}, ~)`
- analysis_dir: saving file path
- 'beh.dat': filename

Save Result files

Saved Analysis files

```
/7/                /8/                /11/
beh.dat            beh.dat            beh.dat
beh_norm.dat       beh_norm.dat       beh_norm.dat
exp_params.dat     exp_params.dat     exp_params.dat
```

\Analysis\~.dat

MATLAB Script

Load multi-session file data

① Load tabular dataset

`tD = load_tabular_dataset_info()`
- tD: used for filtering subset

② Define Meta data for each file

ex)
- `ResultsExt{nResults} = 'beh_'` : Column name tag
- `ResultsHeader{nResults} = 'CELL durTILick durITILick ··· postUnSpd'` : Column name
- `ResultsSummary{nResults} = 'beh.dat'` : file path

used to generate unitKey

③ `LoadPopulationData5Key` : generate aPD by using defined meta data
- aPD: large table of whole loaded data

tD & aPD

MATLAB Work Space

unitKey | beh.dat | beh_norm.dat | exp_params.dat

- unitKey: session# (mid), round# (neural information)
- beh.dat: beh_durTILick = 6
- beh_norm.dat: behn_durTILick = 25

<aPD>

7	1	1	-1	-1
7	2	1	-1	-1
7	3	1	-1	-1
8	1	1	-1	-1
8	1	2	-1	-1
8	2	1	-1	-1
11	1	1	-1	-1
11	3	1	-1	-1
11	2	1	-1	-1

NaN NaN ··· NaN

<tD>

data_tags | datanames

animal	7	8	11
VR_DAY1	m7s1r1	m8s2r1	
VR_DAY2	⋮	⋮	m11s1r1
VR_TWO_MOVING_BARS			
VR_LINEARTRACK			

filter subset

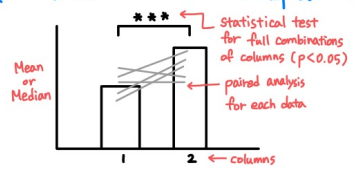
ex) `bV = aPD(:,1)==7` : filter by mid (7)

`bVRD1 = is_uk_member(aPD(:,1:5), get_datanames(tD, {'VR_DAY1'}, ~))`
: filter by tD data_tags ('VR_DAY1')

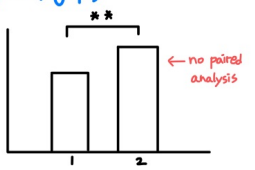
Population Analysis

`[val,grp]=cols2grp()` → val, grp: column vectors

1) `plot_barpair (aPD(bV, [beh_antLick beh_postLick]))`



2) `plot_bargrp (val, grp)`



PSTH files

`[~, psth_1] = plot_timecourse()`
: obtain single-session psth result (psth_1 structure)

`asave_psth(asfullfile('~'), psth_1, data.dataname)`
: Save single-animal psths in one file (.mat)

\Analysis\~.mat

```
/7/                    /8/                    /11/
lick_ab_RewOn.mat      lick_ab_RewOn.mat      lick_ab_RewOn.mat
```

MATLAB Script

① Load tabular dataset

`tD = load_tabular_dataset_info()`

② `all_lick_ab_RewOn = load_psth_files('lick_ab_RewOn.mat', [7, 8, 11], ~)`
- all_lick_ab_RewOn: structure variable
- 'lick_ab_RewOn.mat': Only single filename is available
- [7, 8, 11]: mids of interest

tD & psths

MATLAB Work Space

Field
- m7s10r1 (saved psth structure)
- m7s11r1
- ⋮
- m8s13r1
- m8s14r1
- m8s11r1
- ⋮
- m11s7r1
- m11s10r1
- ⋮

<all_lick_ab_RewOn (structure)>

data_tags | datanames

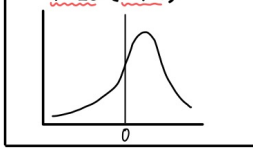
animal	7	8	11
VR_DAY1	m7s1r1	m8s2r1	
VR_DAY2	⋮	⋮	m11s1r1
VR_TWO_MOVING_BARS			
VR_LINEARTRACK			

<tD>

ex) `filt_1 = filter_psth(all_lick_ab_RewOn, [8 NaN NaN NaN NaN])`
: filter by mid (8)
- [8 NaN NaN NaN NaN]: unit Key

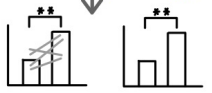
`filt_2 = filter_psth(all_lick_ab_RewOn, get_datanames(tD, 'VR_DAY1'))`
: filter by tD data_tags ('VR_DAY1')

1) `plot_mpsths(filt_2)`
: plot mean of multi-session psths



2) `mpsths2rate()`
: calculate mean of multiple psths within a specific time window

→ `plot_barpair` `plot_bargrp`



3) `plot_mpsths_table (tD, {'VR_DAY1','~'}, lick, neuron)`
: plot psths in a tabular manner

